

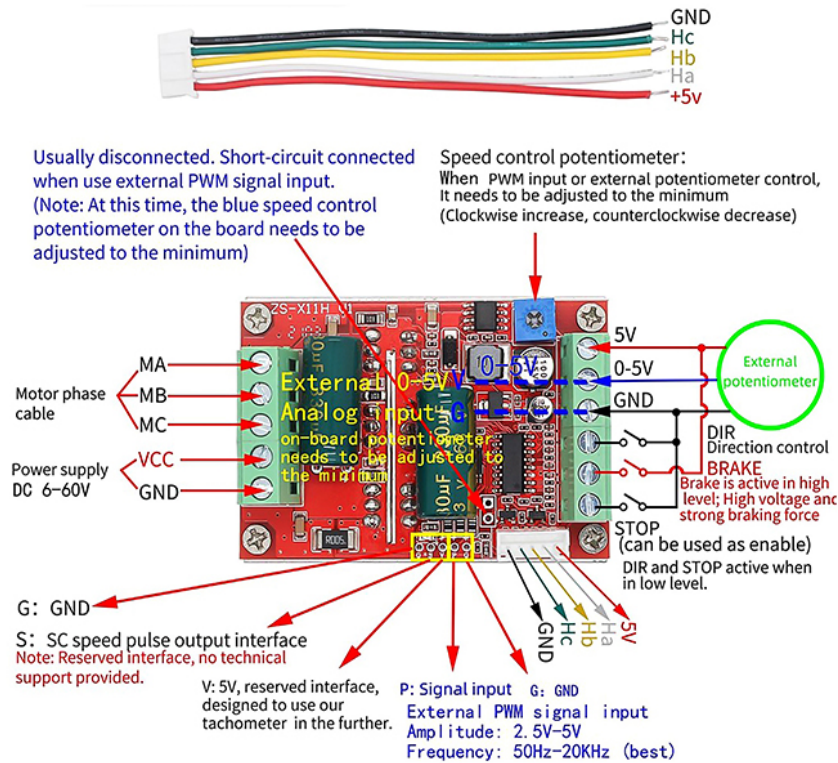
- Product Name: DC Brushless Motor; Shaft Diameter: 5mm, Number of Poles: 8; Number of Phase: 3;
- Cable: Orange: U; Grey: V; Brown: W; Red: +5V; Black: GND; Yellow: H1; Green: H2; White: H3;
- The DC Brushless Motor Made of 100% pure copper coils, the life of the DC brushless motor is 5 times that of the traditional motor, reducing maintenance and replacement costs;

RioRand DC Brushless Controller-350W

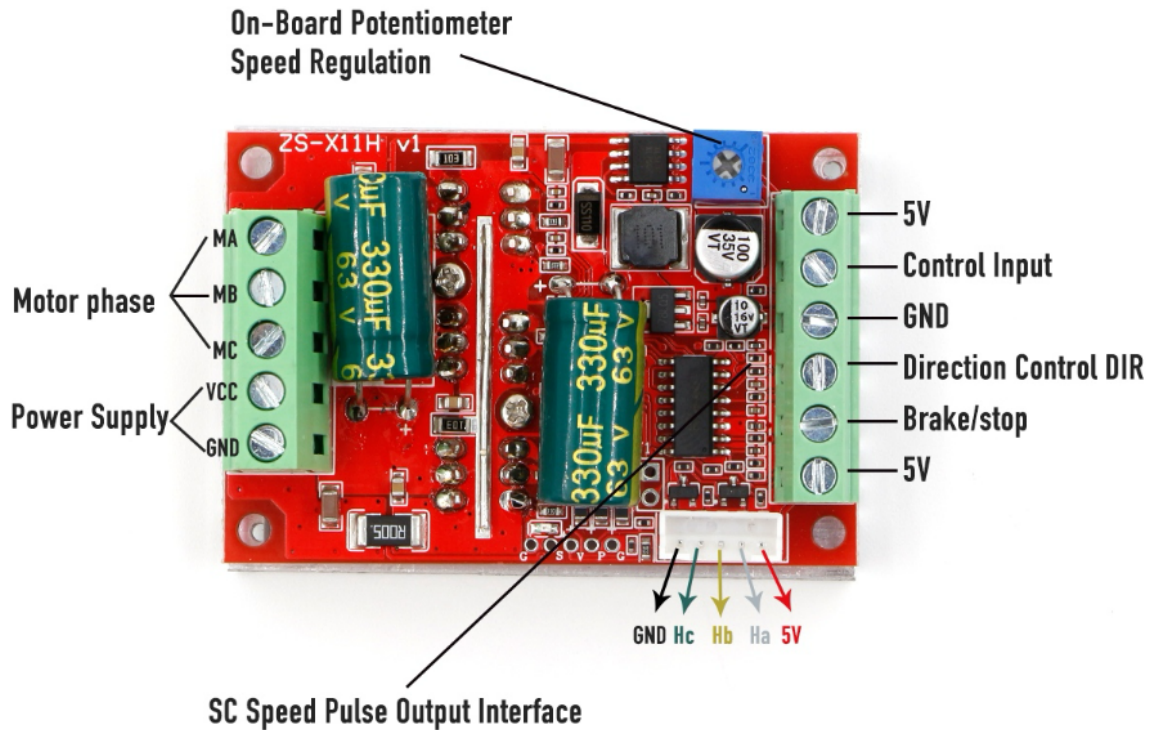
Product Name: 350W Brushless DC Motor Driver with Hall Sensor

Power Supply: DC	Input Voltage: 60V
Dimensions: Approx. 2.48× 1.77× 1.22inches	Weight: Approx. 0.17 lbs
Max Output Current: 20A	Continuous Output Current: 16A
Operating Current: Rated 16A, Peak 20A	Phases: 3-phase
Operating Voltage: 6-60V (60V max; use with safety margin)	Overcurrent Protection: Included
Power: ated 200-300W, Peak 350W (use with safety margin; enhance heat dissipation at high power)	
Compatible Motor: 3-phase brushless DC motors with Hall sensor	

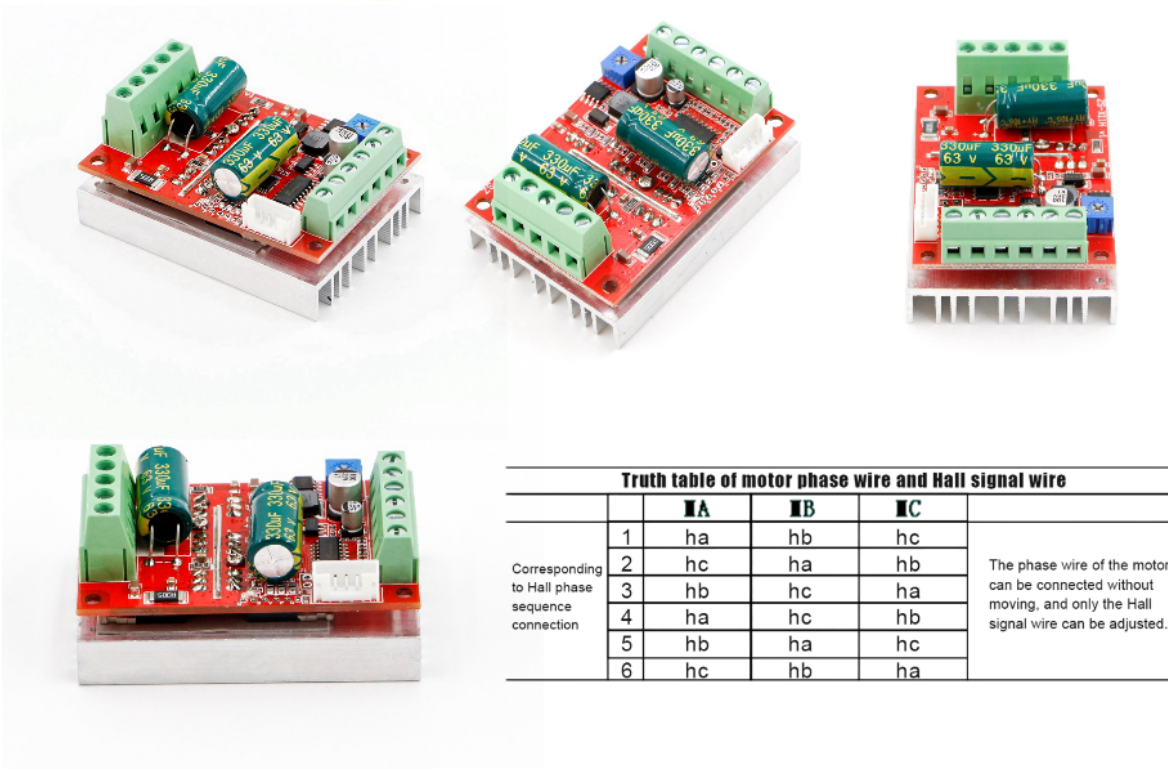
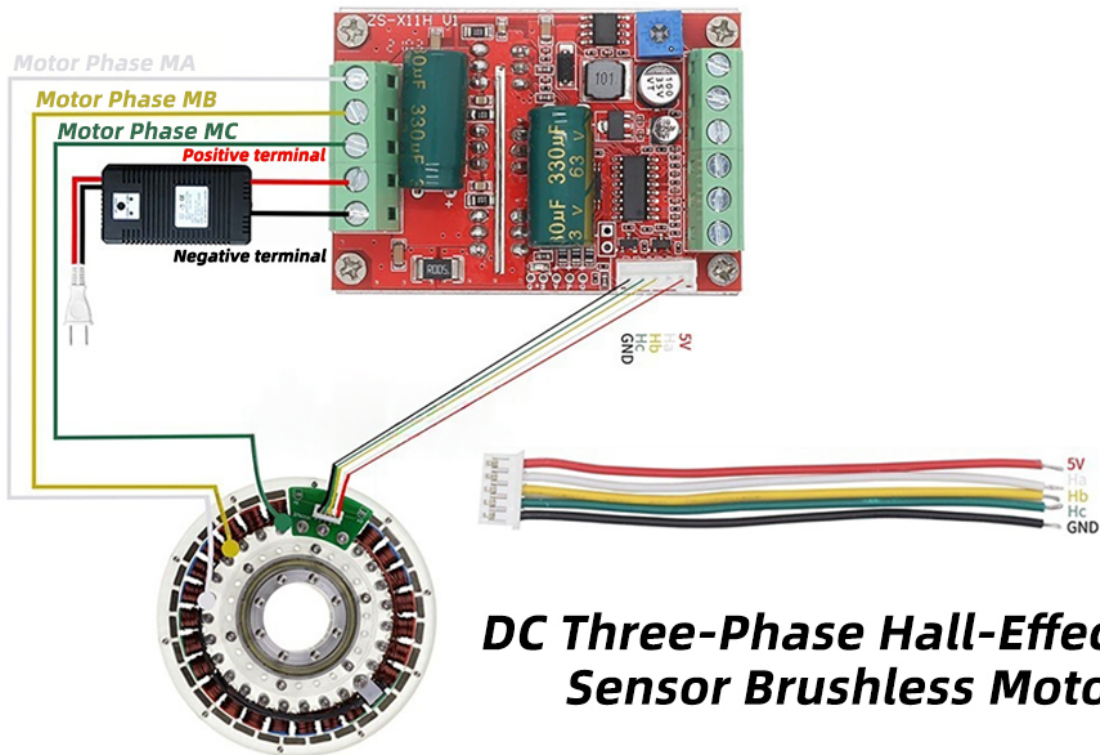
BLDC wide voltage 6-60V, high power 350W 3-phase brushless DC controller with Hall sensor. Supports PLC 0-5V analog control and PWM control (2.5-5V amplitude).



- Special Notes on External Signal Input:
- 1. When connecting external PWM signals via the main board's P and G ports: Set the on-board potentiometer to minimum, and short-circuit the on-board shorting terminals (see wiring diagram above for reference). Signal amplitude: 2.5-5V; frequency: 50Hz-20kHz.
- 2. For external 0-5V analog input: Connect via the main board's 2 terminal ports (set the on-board potentiometer to minimum; see dashed lines in the wiring diagram above for reference).
- 3. For external potentiometer speed control: Connect to the main board's 3 corresponding terminal ports (set the on-board potentiometer to minimum; see potentiometer wiring diagram above for reference).
- 4. Forward/reverse and stop functions: Activate via a switch connected to ground (0V) — active low. External MCU can also send low-level signals. For braking: Activate via a switch connected to 5V (front terminal row) — active high. All front-row terminals accept low-voltage signals directly input to the MCU; do not expose them to high voltage or high current.



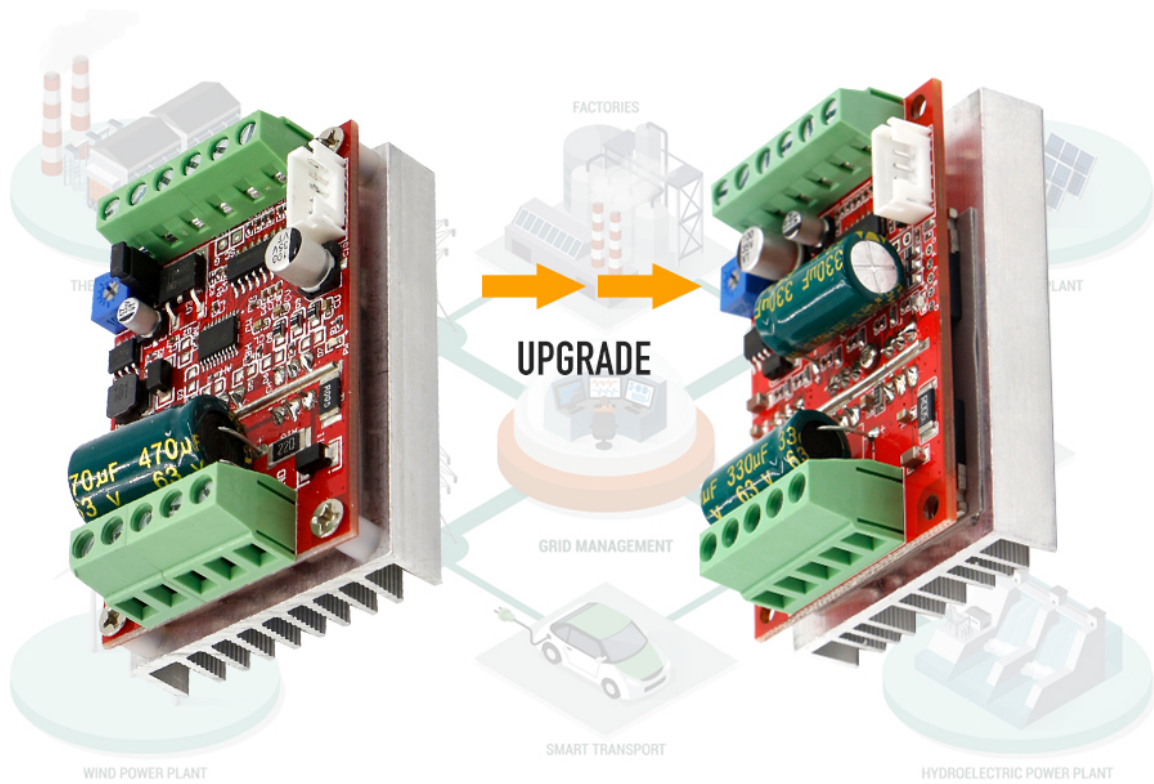
- Interface Description:
- 1.MA, MB, MC (phase outputs) connect to motor
- 2.5V, GND – 5V power from the main board (can supply external devices, max 50mA current)
- 3.VCC, GND – main power supply (external DC power required)
- 4.SC – speed pulse signal output (interface provided, no technical support)
- 5.DIR – direction control (forward/reverse), active low (can connect external switch to GND)
- 6.STOP – stop control, active low (can connect external switch to GND)
- 7.BRAKE – brake control, active high (can connect external switch to 5V)
- 8.Speed control – speed signal input (on-board potentiometer included; also accepts external 0-5V analog or PWM signals)
- 9.Ha, Hb, Hc, +5V, GND – Hall signal & power input (most Hall-equipped motors have these 5 wires)



- After receiving the driver board, first confirm whether its interface is compatible with your brushless motor's interface.
- 1.The brushless motor has 5 Hall wires or a Hall interface. Among them, 2 are Hall power supply wires, and 3 are Hall signal wires—you must distinguish these, especially

the Hall power supply wires (usually red and black). The 3 Hall signal wires are generally labeled a(Ha), b(Hb), c(Hc); the driver board also has 3 ports marked Ha, Hb, Hc (or similar). Connect them one-to-one correctly.

- 2. The motor has 3 thick phase wires, and the driver board also has 3 phase wire interfaces—these are typically labeled MA(U), MB(V), MC(W) (or similar). Connect them one-to-one correctly. Once wired accurately, apply the main power, and turn the potentiometer to around 20%—the motor should run smoothly. (If the motor fails to work normally after connecting interfaces according to labels, follow Step 3 below to troubleshoot unclear interface definitions.)
- 3. If you're unsure about the motor's phase wire/Hall wire definitions, or if the motor doesn't work after incorrect wiring, follow these steps to adjust the Hall wire sequence gradually:
 - First, connect the motor's 3 phase wires to the driver board's phase wire interfaces arbitrarily.
 - For the motor's 5 Hall wires: Connect the 3 Hall signal wires to the driver board's 3 Hall signal ports arbitrarily. The 2 Hall power supply wires must be connected correctly (the main board provides 5V power; one Hall lead is red for 5V, the other is black for ground—ensure these match the motor's Hall power supply wires).
 - After verifying correct connections, test with low voltage (7-12V) and low current (1-3A) (use a constant-current power supply, low-power power supply with a current-limiting resistor, etc., if available). Turn the board's built-in potentiometer to ~20% throttle. If the motor doesn't run or acts abnormally, cut power immediately.
 - Next, adjust the sequence of the motor's 3 phase wires (the driver board's phase interfaces use terminal blocks for easy wiring, so adjusting the motor's phase wires is simpler—adjusting the 3 Hall signal wires instead works the same if the motor's phase wires are fixed). Repeat until the motor runs smoothly when powered on—this means wiring is correct.
 - There are 6 possible phase wire sequences. For example, if the current sequence is ABC, the other options are ACB, BAC, BCA, CAB, CBA. Test power after each adjustment—only 1 sequence is correct.
- 4. If wiring is incorrect and the motor malfunctions, never test with high current or high voltage—this risks damaging the driver board. When the 3 motor phase wires are correctly sequenced, the motor will run as smoothly as silk once powered on, with stable low-speed startup and high torque. Incorrect phase wire sequencing typically causes the following issues:
 1. a. The motor fails to start (no response) or only jerks once without normal rotation.
 2. b. Minor jitter, difficult startup, faint noise, or requiring manual assistance to start.
 3. c. The motor runs in one direction but not the other, with faint noise.
 4. d. Severe startup jitter, weak power, noise, significantly higher current, and overheating of power transistors.



The PWM brushless DC motor speed controller with Hall sensor support accommodates eight-pole motors and features automatic recognition without requiring any adjustments.